Sales Performance Analysis

ApexPlanet Data Analytics Internship Task 4: Data Storytelling & Statistical Validation Prepared by: R. Purushotham Reddy

Problem Statement

Businesses generate large volumes of sales data every day. Raw data is difficult to interpret and does not directly support decision-making. This project converts raw sales data into meaningful business insights using data cleaning, SQL, Power BI dashboards, and statistical validation.

Project Objectives

• Clean and prepare the sales dataset • Perform exploratory data analysis • Answer business questions using SQL • Build an interactive Power BI dashboard • Validate findings using hypothesis testing

Dataset & Data Preparation

The dataset contains customer, product, sales, quantity, pricing, and location information. Data preparation involved removing duplicates, handling missing values, standardizing formats, and creating new features such as Order Month, Order Year, and Sales Category.

Exploratory Data Analysis

EDA was performed to identify trends, distributions, and relationships. Histograms, correlation analysis, SQL queries, and category-wise analysis helped uncover meaningful business patterns and customer behavior.

Interactive Dashboard

The Power BI dashboard includes Total Revenue, Total Orders, Total Customers, Average Order Value, Monthly Sales Trend, Category Revenue, City Revenue, Product Performance, and interactive slicers for dynamic analysis.

Key Business Insights

• Total Revenue exceeded ■139 Million • Average Order Value: ■139,399.44 • High-value customers contribute significantly to revenue • Interactive filters help identify sales trends across cities and categories.

Hypothesis Testing

H■: Average Order Value ≤ ■10,000 H■: Average Order Value > ■10,000 Method: One-Sample t-Test Results: Sample Size=1000, t=35.863, p=5.37×10■¹■² Decision: Reject H■.

Business Recommendations

Increase marketing in high-performing regions, promote premium products, monitor KPIs regularly using Power BI dashboards, and periodically validate business assumptions using statistical analysis.

Conclusion

This project demonstrates the complete analytics lifecycle—from data preparation and exploration to dashboard development and statistical validation—providing actionable insights for data-driven business decisions.